An Empirical Analysis of Entangled Enhancement in an Autonomous Driving Pipeline

***Abstract*—Entangled enhancement is a defective retraining of a machine learning (ML) model that improves the updated model performance but fails to improve the whole system. No study has measured how often this effect occurs in a driving pipeline. This report empirically investigates entangled enhancement in an autonomous driving pipeline consisting of perception, prediction and planning modules. We examine whether an update that improves the perception model increases the error of the prediction and planning models, whether their adaptation to the errors of the perception model causes the increase, and whether entangled enhancement can be detected without ground truth. We train the prediction and planning models on the ground truth or on the output of the perception model before the update, keep both fixed, and update the perception model 30 times. We observe that the prediction error increases in 23 of 30 updates after training on the output of the perception model, in 11 by more than the change caused by a different random seed alone, and in 3 with statistical significance, against 5, 0 and 0 after ground-truth training. The planning error increases by at most 0.13 m. As a result, we confirm that entangled enhancement occurs in the prediction model through its adaptation to the errors of the perception model. None of nine measures without ground truth identifies the updates with entangled enhancement.**

***Index Terms*—autonomous driving, entangled enhancement, machine learning system, model maintenance, retraining**

I. Introduction

A Machine Learning System (MLS) is often built as a pipeline of ML models sequentially connected, where each model uses the output of the preceding model. A model earlier in the sequence is upstream, and a model later in the sequence is downstream. Retraining an upstream model can improve that model and still fail to improve the whole system. Wang and Machida call this effect entangled enhancement [1], [2]. Their work decides when an MLS should retrain each of its models. This decision needs the probability that a retraining causes entangled enhancement. No study has measured this probability on a real system.

A typical autonomous driving pipeline consists of three modules supported by machine learning (ML) models. The perception module finds the objects around the ego vehicle. The prediction module predicts where each object will move, and the planning module chooses the path of the ego vehicle.

Several studies describe the risk of entangled enhancement. Sculley et al. [3] and Wu et al. [4] show that a change to one model of an MLS can make the whole system worse. Xu et al. [5] show that the prediction error increases when the input of the prediction model comes from a real perception model instead of the ground truth. None of these studies retrains the perception model of a driving pipeline and measures how often the error of the prediction and planning models increases, or which cause produces the increase.

In this report, we empirically investigate whether an update that improves the perception model increases the error of the prediction and planning models, and whether the increase comes from the adaptation of these models to the errors of the perception model. Adaptation means that the prediction and planning models, trained on the output of the perception model, learn to work with the errors of the perception model. We also examine whether entangled enhancement can be detected without ground truth. To separate the cause, we train the prediction and planning models either on the ground truth or on the output of the perception model before the update, keep both fixed, and update the perception model 30 times.

We observe that the prediction error increases in 23 of the 30 updates after training on the output of the perception model, and in 5 after training on the ground truth. As a result, we confirm that entangled enhancement occurs in the prediction model and comes from its adaptation to the errors of the perception model. The planning error increases by at most 0.13 m. None of nine measures without ground truth identifies the updates with entangled enhancement.

II. Background and Related Work

This section summarises the related work. Sculley et al. [3] describe how a change to one model of an MLS changes the behaviour of every model that uses the changed output. Wu et al. [4] show that improving one model of a system can make the whole system worse. In this report, we distinguish two causes of entangled enhancement. First, the downstream models adapt during training to the errors of the upstream model and are disturbed when the retraining changes those errors. Second, the upstream model improves on average but becomes worse on the cases that matter downstream.

The prediction and planning models of this report follow two published models. AutoBot-Ego [6] predicts the path of an object. PlanT [7] plans the path of the ego vehicle from a list of objects. Weng et al. [8] show that a prediction model can learn by attention which of the objects found in different images are the same object. The prediction error increases when the input comes from a real perception model instead of the ground truth [5]. A planning model that receives the state of the ego vehicle can score well on the nuScenes driving data set while ignoring the objects [9], [10].

Rabanser et al. [11] detect a change in the data seen by a model without ground truth from distances between distributions, where a distribution is the pattern of values that

an output takes over many inputs. In this report, we apply such distances to the outputs of the models as a candidate symptom, ~~which is~~ a sign of entangled enhancement ~~that needs no ground truth~~.

III. Methodology

This section defines the target pipeline and the notation, and states the research questions and the hypotheses of this report.

A. Target Pipeline and Notation

This report focuses on an autonomous driving pipeline that consists of a perception module, a prediction module and a planning module. We write C_1 for the perception model, C_2 for the prediction model and C_3 for the planning model. C_1 finds the objects in the image of the front camera. C_2 predicts the path of each object over the next 3 s. C_3 chooses the path of the ego vehicle over the same 3 s. An update is one retraining of C_1 on new data. We write C_1' for the updated C_1.

We write δ_1 for the gain of C_1 in mAP50, the standard accuracy measure for object detection. The prediction error is the minADE, which is the average distance between the predicted path and the true path of an object, taken for the best of the paths that C_2 predicts. The planning error is the distance between the planned position and the position of the human driver at 3 s. We write Δ_2 and Δ_3 for the increase of the prediction error and of the planning error after an update. An update ~~shows~~ entangled enhancement when

$$\delta_1 > 0 \quad \text{and} \quad (\Delta_2 > 0 \text{ or } \Delta_3 > 0). \tag{1}$$

B. Research Questions and Hypotheses

The goal of this report is to confirm whether entangled enhancement occurs in this pipeline, to find its cause, and to test whether it can be detected without ground truth. Each of the three research questions below leads to one hypothesis.

RQ1 asks whether an update that improves C_1 can increase the error of C_2 or C_3. The retraining decisions of Wang and Machida need the probability of this event, which no study has measured in a driving pipeline.

H1 (an update that improves C_1 can increase the downstream error): An update of C_1 can increase the prediction error or the planning error, even when the update makes C_1 more accurate.

RQ2 asks whether the increase comes from the adaptation of C_2 and C_3 to the errors of C_1. Only training on the output of C_1 lets C_2 and C_3 learn these errors, which the update then changes.

H2 (the increase comes from the adaptation to the errors of C_1): The increase is larger when C_2 and C_3 are trained on the output of C_1 than when C_2 and C_3 are trained on the ground truth.

RQ3 asks whether the updates with entangled enhancement can be detected without ground truth, because new data after deployment have no ground truth. A larger change in the output of C_1 or C_2 can cause a larger increase of the error.

H3 (the updates with entangled enhancement change the outputs more): An update with entangled enhancement changes the outputs of C_1 and C_2 more than an update without entangled enhancement. The gain δ_1 alone does not indicate entangled enhancement.

IV. Experimental Setup

This section presents the data set, the workflow of the experiment, and the ML models used for experiments. We trained all models on one Tesla T4 graphics card.

A. Data

We use 350 of the 850 scenes of nuScenes [12], which are driving clips of about 20 s from Boston and Singapore, with the full-size images of the front camera only (4.6 GB). Boston 1 (100 scenes) trains C_1, and Boston 2 (100 scenes) trains C_2 and C_3, with 14 of the scenes kept for validation. Separate scenes give C_2 and C_3 the realistic errors of C_1 [5]. Singapore 1 (100 scenes) is used for the updates, and Singapore 2 (50 scenes, 2,020 images) is used for testing. A keyframe is an image with labelled objects, recorded twice per second. The test set has 1,720 test samples, one for every keyframe with 3 s of future.

B. Workflow

Fig. 1 shows the six stages of the experiment. We split the data into four parts. Boston 1 and Boston 2 train the models, Singapore 1 provides the data for the updates, and Singapore 2 is the test set. We train C_1 with two random seeds, where a random seed is the starting number for the random choices made during training. We train C_2 and C_3 once in two training modes. In GT training, we train C_2 and C_3 on the ground truth (GT). In C_1-output training, we train C_2 and C_3 on the output of C_1. We then test each of the 30 updates in the full pipeline with C_2 and C_3 kept fixed.

C. Update Procedure

Algorithm 1 gives the procedure for one C_1, where ~~GT marks~~ GT training and ~~out marks~~ C_1-output training. The two C_1 trained with seeds 0 and 1 have almost the same accuracy. Replacing one by the other still changes the errors of C_2 and C_3, although neither C_1 is better. We call the largest of these changes the seed variation, τ_2 for the prediction error and τ_3 for the planning error. An increase smaller than the seed variation can come from randomness alone. We count the increases at three levels of evidence. The first level counts any increase. The second level counts an increase above the seed variation. The third level counts an increase whose 95 % bootstrap interval lies above zero. The bootstrap repeats the measurement on test sets drawn with replacement from the test scenes.

D. Updates of the Perception Module

We use YOLOv11n [13] as a perception model (C_1), which is trained for 10 passes over Boston 1 with the random seeds 0 and 1. The two C_1 reach an mAP50 of 0.109 and 0.117 on Singapore 2. In each update, we retrain C_1 on Singapore 1 with one of 15 settings. Nine settings use 100 %, 50 % or 25 % of the images with 10 passes, and six use all images with 5

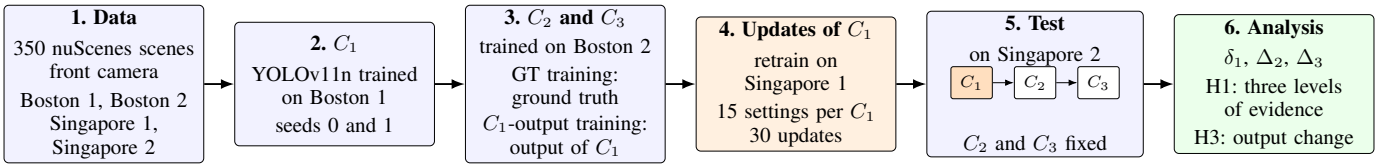


Fig. 1. Workflow of the experiment

Algorithm 1: Update procedure for one C_1

Input : C_1, settings $\mathcal{S}$, Boston 2, Singapore 1, Singapore 2
Output: δ_1, Δ_2, Δ_3 and evidence level per update and training
Train $C_2^{\mathrm{GT}}, C_3^{\mathrm{GT}}$ on the ground truth of Boston 2
Train $C_2^{\mathrm{out}}, C_3^{\mathrm{out}}$ on the output of C_1 on Boston 2
foreach *training* $r \in \{\mathrm{GT}, \mathrm{out}\}$ **do**
  $E_2^r, E_3^r \leftarrow$ errors of $C_1 \rightarrow C_2^r \rightarrow C_3^r$ on Singapore 2
$\tau_2, \tau_3 \leftarrow$ seed variation from the C_1 with the other seed
foreach *setting* $s \in \mathcal{S}$ **do**
  $C_1' \leftarrow$ retrain C_1 on Singapore 1 with s
  $\delta_1 \leftarrow \mathrm{mAP50}(C_1') - \mathrm{mAP50}(C_1)$
  foreach *training* $r \in \{\mathrm{GT}, \mathrm{out}\}$ **do**
    $\Delta_2, \Delta_3 \leftarrow$ errors of $C_1' \rightarrow C_2^r \rightarrow C_3^r$ minus E_2^r, E_3^r
    record the three levels of evidence
  record the output change of C_1 and C_2 for H3

or 20 passes, each with the seeds 1, 2 and 3. The 15 settings applied to both C_1 give 30 updates. The test scenes are never used to choose a model.

E. ML Prediction and Planning Modules

AutoBot-Ego [6] is employed as a prediction model (C_2). For each current object, a transformer reads the objects of the last four keyframes relative to that object and outputs six possible paths of 3 s, without a hand-written rule that links objects across images [8]. For the planning model (C_3), we employ PlanT [7]. Every object is one input with the current position, the most likely path from C_2, the class and the confidence. Two further inputs hold the driving command and the path of the ego vehicle over the last 1.5 s, which we call the ego path. The command is left, straight or right [9], and a recurrent decoder outputs six points of the planned path.

We train both models on Boston 2 with early stopping on the validation scenes and with mirror and rotation augmentation. An input of the ego path can let a planning module ignore the objects [10], and we check whether C_3 uses the objects. Removing the objects from the input of C_3 moves the planned position at 3 s by 0.73 to 1.65 m on average, which is above the threshold of 0.5 m fixed before the experiment.

F. Training and Measurements

We train C_2 and C_3 once with GT training and once with C_1-output training for each C_1, and the trained models stay fixed during all updates. The prediction error is computed over the objects found by C_1 that are matched one to one with a ground truth object of the same class. The outputs of C_2 and C_3 on the ground truth stayed exactly the same before and after every update, which confirms that only the output of C_1 changed. The seed variation is $\tau_2 = 0.112$ m and $\tau_3 = 0.018$ m. For the bootstrap, we draw the 50 test scenes 2,000 times with replacement.

TABLE I
ML MODELS AGAINST SIMPLE BASELINES ON THE TEST SET (M)

Model and input	Model	Baseline
C_2 on the ground truth	0.77	4.48 (stays still)
C_2 on the output of C_1	2.39 to 2.95	5.39 to 5.85 (stays still)
C_3 without the ego path	9.3 to 12.5	9.25 (average path)
C_3 with the ego path	3.70 to 4.16	9.25 (average path)

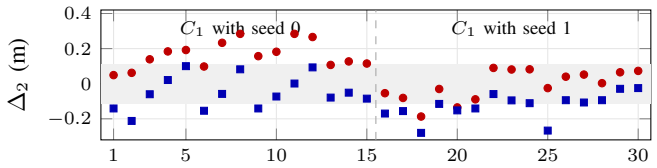


Fig. 2. Δ_2 in the 30 updates. Red circles show C_1-output training, blue squares show GT training, and the grey band shows the seed variation.

V. Results and Discussion

In this section, we first check that C_2 and C_3 work, because a count of updates that increase the error is meaningful only for working modules. We then report and discuss the results. Every update made C_1 more accurate, with δ_1 between 0.048 and 0.118.

A. Validity of the ML Components

We compare C_2 and C_3 with simple baselines on the test set (Table I). C_2 beats the baseline in which every object stays still. The variant of C_3 without the ego path does not beat the baseline that drives the average Boston path for each command. We use the variant with the ego path, which has an error of 3.70 to 4.16 m against 9.25 m for the baseline.

B. Prediction Module

Fig. 2 shows Δ_2 for every update in two training modes. In all 30 updates, Δ_2 after C_1-output training is larger than Δ_2 after GT training. The difference is largest for the updates from the C_1 with seed 0, where Δ_2 after C_1-output training lies between $+0.049$ and $+0.284$ m with a mean of $+0.165$ m.

Table II counts the updates with an increase at the three levels of evidence. We observe that, above seed variation, the prediction error increased in 11 updates after C_1-output training and in none after GT training. All 11 updates start from the C_1 with seed 0. With 50 test scenes, the bootstrap

TABLE II
UPDATES, OUT OF 30, WITH AN INCREASE OF THE ERROR

	Prediction Δ_2		Planning Δ_3	
Level of evidence	C_1**-output**	**GT**	C_1**-output**	**GT**
Any increase	23	5	18	29
Above seed variation	11	0	13	29
Bootstrap interval above zero	3	0	1	10

interval of Δ_2 spans about ± 0.25 m. Only 3 updates after C_1-output training and none after GT training have an interval above zero.

The comparison results characterize the first cause of entangled enhancement. Both versions of C_2 receive the same output of C_1'. Only the version that learned from the output of C_1 becomes worse (Fig. 2).

C. Planning Module

The planning error changed little. The mean Δ_3 lies between -0.006 and $+0.058$ m, and the largest value is 0.13 m against a planning error of about 4 m. The increase appears more often after GT training than after C_1-output training at every level in Table II. Adaptation to the errors of C_1 cannot explain an increase after GT training.

Δ_3 is small because C_3 plans mainly from the ego path. The objects move the planned position at 3 s by only 0.73 to 1.65 m. The more frequent increase after GT training points to the second cause. The C_3 trained on the ground truth receives the predicted paths of C_2 only at test time. A change in the objects that C_1' finds can therefore affect this C_3 more than the C_3 trained on the output of C_1.

D. Retraining C_2 and C_3 on the Output of C_1'

We test whether retraining C_2 and C_3 together with C_1 lowers the prediction error and the planning error below their values before the update. We retrain C_2 and C_3 on the output of C_1' for two further updates, one from each C_1 with all images and 10 passes. After C_1-output training, the prediction error after the update is 2.75 and 2.61 m, and the planning error is 4.14 and 4.00 m. After the retraining, the prediction error is 2.39 and 2.45 m, and the planning error is 3.71 and 3.70 m.

The retraining lowers the prediction error by 0.36 and 0.16 m and the planning error by 0.44 and 0.30 m. The errors also fall below the errors before the update (2.62 and 2.66 m, 4.16 and 4.00 m), and the update becomes an improvement of the whole pipeline. This result supports the retraining of downstream models that Wang and Machida model [1], [2]. ~~This retraining has a cost. A symptom of entangled enhancement is therefore useful.~~

E. Symptom of Entangled Enhancement

Nine measures of output change need no ground truth, seven on the output of C_1 and two on the output of C_2. The rank correlation with Δ_2 is close to 1 when a larger measure always comes with a larger Δ_2. The AUROC is the probability that an update with $\Delta_2 > 0$ has a larger measure than an update with $\Delta_2 \leq 0$. Over all 30 updates, the shift of the predicted position of the same object has the highest rank correlation ($+0.68$, AUROC 0.81). For each C_1 separately, the rank correlation is only $+0.29$ and -0.28. The high value over all updates comes from the difference between the two C_1. The gain δ_1 has a rank correlation of $+0.04$. No measure consistently separates the updates that increase the prediction error from the other updates. H3 is therefore not supported.

VI. CONCLUSION

In this report, we ~~updated the perception module 30 times~~. We observe that the prediction module became worse more often and by more after C_1-output training than after GT training. This result supports H1 and H2 for the prediction model and points to adaptation to the errors of the perception model before the update as the cause. The planning error changed little. No measure of output change that needs no ground truth identified the updates that increased the prediction error. H3 is therefore not supported. As future work, we develop a method that detects entangled enhancement after an update of the perception module, before the update is used and before ground truth for the new data exists.

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